Technical University of Munich
School of Computation, Information, and Technology
Department of Computer Science, SS 2026

Operations Research

Prof. Dr. Martin Bichler,
Halil Bayrak, Janik Bürgermeister, Teodora Dobos,
Jan Höhener, Abheek Gosh

Solution Demo exercises, Sheet 2

Demo Exercises

Exercise D2. 1. *Storage*

Problem

A storage room is to be equipped with shelves. The room has a total of $72m^2$ of floor space available for shelves. There are two types of shelves to choose from. The first shelf costs 10€, requires $6m^2$ of floor space, and provides $8m^3$ of storage capacity. The second shelf costs 20€, requires $8m^2$ of floor space, and provides $12m^3$ of storage capacity. How many shelves of each type would you buy to maximize the storage capacity if your budget is 140€? Determine the optimal solution using the Simplex algorithm.

Solution

1. Setting up the Linear Program

Variables:

x_1	Number of shelves of the 1st type purchased
x_2	Number of shelves of the 2nd type purchased

Model:

Maximize

$$8x_1 + 12x_2 = z$$

subject to

$$\begin{aligned} 10x_1 + 20x_2 &\leq 140 \quad (Cost) \\ 6x_1 + 8x_2 &\leq 72 \,(Floor\ space) \\ x_1, x_2 &\geq 0 \end{aligned}$$

2. Simplex Algorithm

LP in Standard Form:

Maximize

$$\begin{array}{rcl} 8x_1 + 12x_2 & = & z \\ \text{subject to} & & \\ 10x_1 + 20x_2 + x_3 & = & 140 \\ 6x_1 + 8x_2 + x_4 & = & 72 \\ x_1, x_2, x_3, x_4 & \geq & 0 \end{array}$$

Solution:

Initial basis: x_3, x_4

Base	z	x_1	x_2	x_3	x_4	Res.
z	1	-8	-12			0
x_3		10	$\boxed{20}$	1		140
x_4		6	8		1	72

Basis swap: $x_2 \leftrightarrow x_3$

Base	z	x_1	x_2	x_3	x_4	Res.
z	1	-2		$3/5$		84
x_2		$1/2$	1	$1/20$		7
x_4		$\boxed{2}$		$-2/5$	1	16

Basis swap: $x_1 \leftrightarrow x_4$

Base	z	x_1	x_2	x_3	x_4	Res.
z	1			$1/5$	1	100
x_2			1	$3/20$	$-1/4$	3
x_1		1		$-1/5$	$1/2$	8

The stopping rule is satisfied (only non-negative entries in the z-row). The optimal solution is given by $x^* = (8, 3)$ with the corresponding objective function value $z_{\max} = 100$.

Exercise D2. 2. *Simplex-Algorithmus*

Problem

Consider the following Linear Program (LP):

$$\begin{array}{llcl} \text{Max} & 2x_1 - x_2 & & \\ \text{s.t.} & -x_1 + x_2 & \geq & 2 \\ & x_1 + x_2 & \leq & 4 \\ & x & \in & \mathbb{R}^2_{\geq 0} \end{array}$$

a) Solve the LP graphically.

b) Transform the LP into standard form.

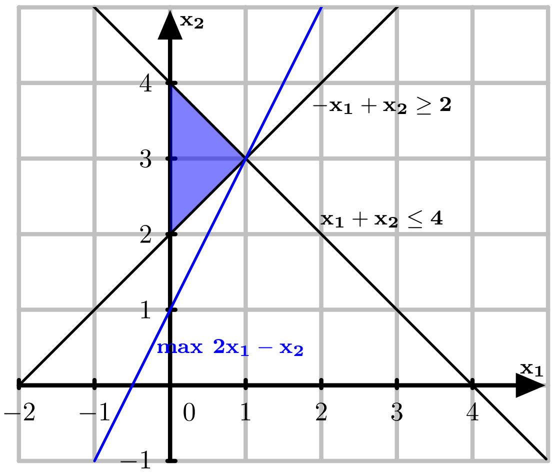
c) Is the LP in canonical form? Justify.

d) Compute an optimal basic solution for the LP using the Simplex algorithm, utilizing the Big-M method.

e) Mark the feasible basic solutions found in part (d) on your graphical solution from part (a).

Solution

a) Graphical Solution:



b)

$$\begin{array}{lrcl} \text{Max} & 2x_1 - x_2 & & \\ \text{s.t.} & -x_1 + x_2 - x_3 & = & 2 \\ & x_1 + x_2 + x_4 & = & 4 \\ & x & \in & \mathbb{R}^4_{\geq 0} \end{array}$$

c) The LP is not in canonical form since the first constraint lacks a basic variable.

d) We need an additional artificial variable $w_1 \geq 0$ for the first constraint.

$$\begin{array}{lrcl} \text{Max} & 2x_1 - x_2 - Mw_1 = z & & \\ \text{s.t.} & -x_1 + x_2 - x_3 + w_1 & = & 2 \\ & x_1 + x_2 + x_4 & = & 4 \\ & x & \in & \mathbb{R}^4_{\geq 0} \\ & w_1 & \geq & 0 \end{array}$$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-2	1			M	0
w_1	-1	1	-1		1	2
x_4	1	1		1		4

The tableau needs to be transformed into canonical form first. For this purpose, variable w_1 must be eliminated from the objective function.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	$-2+M$	$1-M$	M			$-2M$
w_1	-1	$\boxed{1}$	-1		1	2
x_4	1	1		1		4

Now, Phase 2 of the Simplex algorithm is executed.

Basic variable exchange: $w_1 \leftrightarrow x_2$

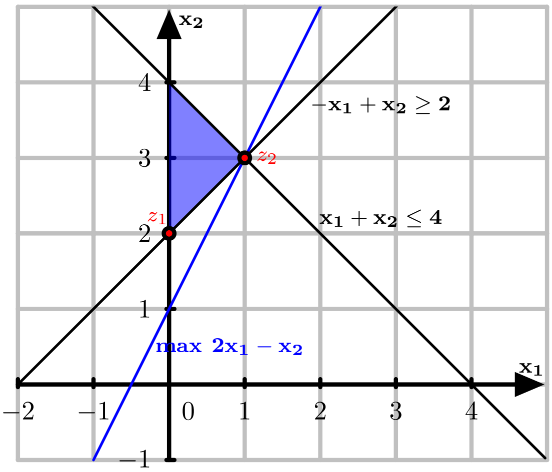
Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	-1		1		$-1+M$	-2
x_2	-1	1	-1		1	2
x_4	$\boxed{2}$		1	1	-1	2

Basic variable exchange: $x_4 \leftrightarrow x_1$

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z			1.5	0.5	$-1.5 + M$	-1
x_2		1	-0.5	0.5	0.5	3
x_1	1		0.5	0.5	-0.5	1

The stopping criterion is fulfilled (only non-negative entries in the objective row). The optimal solution is $x^* = (1, 3)$ with an associated objective value $z_{\max} = -1$.

e) The two feasible basic solutions are $z_1 = (0, 2)$ and $z_2 = (1, 3)$.



Exercise D2. 3. *Simplex-Algorithmus*

Problem

Solve the following linear optimization problem using the Simplex algorithm:

Max

$$4x_1 + x_2 = z$$

s.t.

$$x_1 + x_2 \leq 1$$
$$x_2 \leq 1$$
$$x_1, x_2 \in \mathbb{R}$$

Solution

The variables x_1 and x_2 are unbounded. However, to still be able to apply the Simplex algorithm, x_1 is replaced by $x_1^+ - x_1^-$ and x_2 by $x_2^+ - x_2^-$, where:

$$x_1^+, x_1^-, x_2^+, x_2^- \geq 0$$

Model

Max

$$4x_1^+ - 4x_1^- + x_2^+ - x_2^- = z$$

s.t.

$$\begin{aligned} x_1^+ - x_1^- + x_2^+ - x_2^- &\le 1 \\ x_2^+ - x_2^- &\le 1 \\ x_1^+, x_1^-, x_2^+, x_2^- &\ge 0 \end{aligned}$$

Standard Form:

Max

$$4x_1^+ - 4x_1^- + x_2^+ - x_2^- = z$$

s.t.

$$\begin{aligned} x_1^+ - x_1^- + x_2^+ - x_2^- + x_3 &= 1 \\ x_2^+ - x_2^- + x_4 &= 1 \\ x_1^+, x_1^-, x_2^+, x_2^-, x_3, x_4 &\ge 0 \end{aligned}$$

Initial Basis x_3, x_4

Base	Coefficients							Res.
	z	x_1^+	x_1^-	x_2^+	x_2^-	x_3	x_4	
z	1	-4	4	-1	1			0
x_3	0	$\boxed{1}$	-1	1	-1	1	0	1
x_4		0		1	-1		1	1

Basis Exchange: $x_3 \leftrightarrow x_1^+$

Base	Coefficients							Res.
	z	x_1^+	x_1^-	x_2^+	x_2^-	x_3	x_4	
z	1			3	-3	4		4
x_1^+		1	-1	1	-1	1		1
x_4				1	-1		1	1

The objective function coefficient at x_2^- is $-3 < 0$, and all other coefficients in the x_2^- column are negative. Thus, the objective function value is unbounded, and the algorithm terminates.